

Operadic canonicity: what grafting proves, and where the mounting image is too small

Draft (2026-08-10). Desk report for the operadic-canonicity track. This document makes no design decision. It tests whether the mounting operad removes the qualification from `campbell-renyi.md` §8.

Claim labels used throughout:

- **Verified** means checked against a cited primary source or an already mechanized fact in this repository.
- **Desk-proved** means a complete finite derivation is given here but is not yet mechanized.
- **Conjectured** means a plausible extension whose proof is not supplied.
- **Wall** means the target does not follow in the stated representation or without an additional hypothesis.

1. Verdict first

Desk-proved, repair. Mountings do form an operad under grafting. A mount path is a finite bit word; substituting paths below a mounted head concatenates words. Consequently depths add and cylinder masses multiply:

$$d_{ij} = d_i + e_{ij}, \quad 2^{-d_{ij}} = 2^{-d_i} 2^{-e_{ij}}.$$

This is exactly the machine-native composition law proposed in the track.

Wall, representation. The resulting Kraft-mass operad is *not* the dyadic-rational suboperad of the simplex operad. A head mounted at one path has one mass 2^{-d_i} , not an arbitrary dyadic rational. Incomplete mountings have total mass at most one rather than exactly one. Even after restricting to complete mountings, the fixed-arity image is not dense: a complete two-head mounting has mass vector only $(1/2, 1/2)$. Thus continuity cannot extend a law from physical mounting masses to a whole simplex. This is the precise wall in the proposed Bradley/Faddeev import.

Desk-proved, repair. Put $t > 0$ and $\alpha = 1/(1+t)$. Campbell's continuously optimal endpoint moment is

$$M_t(p) = 2^{tH_\alpha(p)} = \left(\sum_i p_i^\alpha \right)^{1/\alpha}.$$

The quantity satisfying the proposed grafting law is therefore not M_t itself but its α th power, the partition functional

$$Z_\alpha(p) := M_t(p)^\alpha = 2^{(1-\alpha)H_\alpha(p)} = \sum_i p_i^\alpha.$$

For $p \circ (q_i) = (p_i q_{ij})_{i,j}$,

$$Z_\alpha(p \circ (q_i)) = \sum_i p_i^\alpha Z_\alpha(q_i).$$

This is exact in the continuous Kraft relaxation. The exact physical statement for integer paths is the unoptimized fixed-mounting grafting identity; integer optima retain Campbell's rounding gap.

Desk-proved, repair. At $\alpha = 1$, Z_1 is the constant one, so its pointwise limit is not the Shannon chain rule. The chain rule is the *derivative* of the partition law at $\alpha = 1$:

$$H(p \circ (q_i)) = H(p) + \sum_i p_i H(q_i).$$

Equivalently, the Tsallis normalization of $Z_\alpha - 1$ converges to Shannon entropy and obeys the p_i^α -deformed derivation law before taking the limit.

Verified, import rather than reproof. The deformed uniqueness theorem already exists. Baez, Fritz, and Leinster's Theorem 7, using Furuichi's generalized Faddeev theorem, characterizes Tsallis entropy by continuity, functoriality, and degree- α compatibility with convex combinations. In operadic notation its strong-additivity weight is exactly p_i^α . Rényi entropy is the monotone logarithmic transform of the same partition functional. Bradley supplies the topological-operad derivation framework for the Shannon case, but her theorem says that evaluation at zero of any such derivation is a Shannon multiple; it does not say that every entire derivation is the constant Shannon derivation.

Desk-proved, conditional converse. Two clean hypotheses do force the machine schedule to be exponential:

1. if a positive affine normalization of endpoint cost is multiplicative under depth addition, then it is q^d on $d \in \mathbb{N}$;
2. in the differentiable continuous, separable Kraft relaxation, if every optimizer is an escort distribution of one fixed order α , then endpoint cost is $A2^{td} + B$, with $t = (1 - \alpha)/\alpha$.

These results pin the schedule *within the stated ansatz*. The bare physical mounting operad does not supply the missing density or the hypotheses that connect an arbitrary schedule to a simplex functional.

Desk-proved, recommended workaround. Density is unnecessary if workloads and mountings are kept as the two factors of a scheduled-mounting operad $\Delta \times \mathcal{M}$. Local factorization of a separable endpoint observable on that paired operad, tested already on unary paths, forces $g(d + e) = g(d)g(e)$ and hence $g(d) = q^d$. Fiberwise Campbell optimization then induces the escort/Tsallis/Rényi law on the workload factor. This preserves the literal one-path-per-head machine representation and pins the classification under one explicit operadic locality axiom.

2. The mounting operad, precisely

Let B^* be the finite words in $B = \{0, 1\}$, including the empty word ϵ . Write $u \preceq v$ when u is a prefix of v . Two words are prefix-incomparable when neither prefixes the other.

2.1 Operations and grafting

Definition. For $n \geq 1$, the arity- n mounting space $\mathcal{M}(n)$ consists of ordered tuples

$$u = (u_1, \dots, u_n) \in (B^*)^n$$

whose entries are pairwise prefix-incomparable. We use the reduced convention: there are no nullary operations. The symmetric group acts by permuting entries. Given $u \in \mathcal{M}(n)$ and $v^i = (v_1^i, \dots, v_{m_i}^i) \in \mathcal{M}(m_i)$, define simultaneous grafting by

$$u \circ (v^1, \dots, v^n) := (u_i v_j^i)_{1 \leq i \leq n, 1 \leq j \leq m_i},$$

where juxtaposition is word concatenation. The unit is (ϵ) .

Desk-proved. Grafting preserves prefix-incomparability. If two grafted words have different outer indices $i \neq i'$, a prefix relation between them would imply a prefix relation between u_i and $u_{i'}$. If they have the same outer index, canceling the common prefix u_i would give a prefix relation between two distinct entries of v^i . Both contradict the hypotheses. Associativity is associativity of word concatenation, the empty word gives the unit laws, and permutation compatibility is immediate. Hence $\mathcal{M}$ is a symmetric operad.

Verified. This definition matches the repository's mechanized Mounting: one path per head, prescribed path length, injectivity, and pairwise prefix-incomparability. The theorem `Adic.Dyadic.kraft_iff_mounting` proves that this is equivalent to the repository's all-natural Kraft condition.

2.2 Depths and Kraft masses

For $u \in \mathcal{M}(n)$ define

$$d_i(u) = |u_i|, \quad \mu_i(u) = 2^{-|u_i|}.$$

Desk-proved. Grafting gives

$$d_{ij}(u \circ v) = d_i(u) + d_j(v^i), \quad \mu_{ij}(u \circ v) = \mu_i(u) \mu_j(v^i).$$

The first identity is length-additivity under concatenation; exponentiation gives the second.

Let

$$\overline{\Delta}(n) = \left\{ x \in \mathbb{R}_{\geq 0}^n : \sum_i x_i \leq 1 \right\}.$$

It is a subprobability operad under $x \circ (y^i) = (x_i y_j^i)_{i,j}$, since

$$\sum_{i,j} x_i y_j^i = \sum_i x_i \sum_j y_j^i \leq \sum_i x_i \leq 1.$$

Desk-proved. Kraft's inequality says that $\mu : \mathcal{M} \rightarrow \overline{\Delta}$ is an operad map. The complete mountings

$$\mathcal{M}_=(n) = \left\{ u \in \mathcal{M}(n) : \sum_i 2^{-|u_i|} = 1 \right\}$$

form a suboperad, and their masses land in the ordinary probability simplex.

2.3 The image is smaller than the dyadic simplex

Define the physical mass image

$$P_2(n) = \left\{ (2^{-d_1}, \dots, 2^{-d_n}) : d_i \in \mathbb{N}, \sum_i 2^{-d_i} \leq 1 \right\}.$$

Its complete part uses equality. Every coordinate is dyadic rational, but the converse is false: for example $3/8$ is dyadic rational and is not 2^{-d} .

Desk-proved. A complete positive two-coordinate physical mass vector is only $(1/2, 1/2)$. Suppose without loss of generality $a \leq b$ and

$$2^{-a} + 2^{-b} = 1.$$

Multiplying by 2^b gives $2^{b-a} + 1 = 2^b$. The left side is odd unless $a = b$; hence $a = b$, and then $2^{1-a} = 1$, so $a = b = 1$.

Wall. Thus the complete arity-two physical mass image is a singleton and is not dense in the line segment $\Delta(2)$. Allowing incomplete mountings does not help: each coordinate still lies in the discrete set $\{1, 1/2, 1/4, \dots\}$, whose only accumulation point is zero.

The genuinely dyadic simplex

$$D_2(n) = \{p \in \Delta(n) : p_i \in \mathbb{Z}[1/2] \text{ for every } i\}$$

is dense and is closed under simplex-operad composition, but it is not the mass image of one-path-per-head mountings. Realizing $k_i/2^N$ physically requires giving head i a union of k_i depth- N cylinders. That changes a mount slot from one connected subtree to a generally disconnected forest and destroys its single scalar distance.

3. Functionals and derivations without hidden subtraction

There are two different inputs in the machine story:

- a **workload vector** $p = (p_i)$ records head frequencies;
- a **mounting mass vector** $r = (2^{-d_i})$ records allocated fast-state cylinders.

They coincide only at the Shannon continuous optimum. At the Campbell optimum, r is the escort of p , not p itself. Keeping these objects separate is load-bearing.

3.1 Algebra-level functionals

Definition. An algebra-level functional is a symmetric family $F_n : \Delta(n) \rightarrow A$, where A is an ordered commutative additive monoid or a real vector space. It is *mass-local on mountings* when its value on a complete mounting factors through the mass map μ .

Let $w : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ be a scalar weight system.

Definition. A w -derivation in simultaneous-composition form is a family $S_n : \Delta(n) \rightarrow \mathbb{R}$ satisfying

$$S(p \circ (q_i)) = S(p) + \sum_i w(p_i) S(q_i).$$

The strict Shannon derivation has $w(x) = x$. The α -deformed derivation has $w(x) = x^\alpha$.

This is the scalar shadow of Bradley's operadic bimodule derivation. It is also Faddeev's strong additivity law. It states composition directly and uses no coboundary subtraction.

For $0 < \alpha < 1$, define the positive partition functional and Tsallis normalization

$$Z_\alpha(p) = \sum_i p_i^\alpha, \quad S_\alpha(p) = \frac{Z_\alpha(p) - 1}{1 - \alpha}.$$

Desk-proved, exchange form. The deformed law can be written entirely with positive quantities as

$$Z_\alpha(p \circ (q_i)) = \sum_i p_i^\alpha Z_\alpha(q_i).$$

Substituting $Z_\alpha = 1 + (1 - \alpha)S_\alpha$ and using $\sum_i p_i^\alpha = Z_\alpha(p)$ gives

$$S_\alpha(p \circ (q_i)) = S_\alpha(p) + \sum_i p_i^\alpha S_\alpha(q_i).$$

Equation (P) is the exchange-form-friendly version: it avoids both subtraction and division, just as AIP-5 recommends rearranging cost/potential identities into a positive equality.

3.2 Fixed-mounting cost is already operadic

Let $g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ be an endpoint price and define

$$E_g(p, u) = \sum_i p_i g(d_i(u)).$$

Desk-proved. For arbitrary g ,

$$E_g(p \circ (q_i), u \circ (v^i)) = \sum_{i,j} p_i q_j^i g(d_i + e_j^i).$$

If $g(d) = d$, (F) becomes the strict law

$$E_g(p \circ q, u \circ v) = E_g(p, u) + \sum_i p_i E_g(q_i, v^i).$$

If $g(d) = 2^{td}$, it becomes the exact deformed fixed-mounting law

$$E_t(p \circ q, u \circ v) = \sum_i p_i 2^{td_i} E_t(q_i, v^i).$$

No relaxation or optimality is used here. This is the strongest exact law on literal integer mountings.

4. Forward direction: Campbell under grafting

Fix $t > 0$ and put

$$\alpha = \frac{1}{1+t}, \quad t = \frac{1-\alpha}{\alpha}.$$

Let p have positive coordinates. In the continuous Kraft relaxation, use shares $r_i = 2^{-d_i} > 0$ with $\sum_i r_i = 1$. The endpoint moment is

$$E_t(p, r) = \sum_i p_i r_i^{-t}.$$

4.1 Escort optimum

Desk-proved. Define

$$Z_\alpha(p) = \sum_i p_i^\alpha, \quad r_i^* = \frac{p_i^\alpha}{Z_\alpha(p)}.$$

Then

$$\begin{aligned} E_t(p, r^*) &= \sum_i p_i \left(\frac{p_i^\alpha}{Z_\alpha(p)} \right)^{-t} \\ &= Z_\alpha(p)^t \sum_i p_i^{1-\alpha t} \\ &= Z_\alpha(p)^{1+t} = Z_\alpha(p)^{1/\alpha}, \end{aligned}$$

because $1 - \alpha t = \alpha$ and $1 + t = 1/\alpha$.

For any other r with $\sum_i r_i = 1$, Hölder with conjugate exponents $1/\alpha$ and $1/(1-\alpha)$ gives

$$\begin{aligned} Z_\alpha(p) &= \sum_i (p_i r_i^{-t})^\alpha r_i^{1-\alpha} \\ &\leq \left(\sum_i p_i r_i^{-t} \right)^\alpha \left(\sum_i r_i \right)^{1-\alpha} = E_t(p, r)^\alpha. \end{aligned}$$

Thus r^* is optimal and

$$M_t(p) := \min_r E_t(p, r) = Z_\alpha(p)^{1/\alpha} = 2^{tH_\alpha(p)}.$$

Zero-probability coordinates can be deleted before the relaxation; literal finite mountings may place them in unused deep slots afterward.

4.2 Exact value law and hierarchical optimizer

Let $p \in \Delta(n)$ and $q_i \in \Delta(m_i)$. Write $P_{ij} = p_i q_j^i$ and $Z_i = Z_\alpha(q_i)$.

Desk-proved. Directly,

$$\begin{aligned} Z_\alpha(P) &= \sum_{i,j} (p_i q_j^i)^\alpha \\ &= \sum_i p_i^\alpha \sum_j (q_j^i)^\alpha \\ &= \sum_i p_i^\alpha Z_\alpha(q_i). \end{aligned}$$

Combining (C) and (G) gives the exact Campbell value law

$$M_t(P)^\alpha = \sum_i p_i^\alpha M_t(q_i)^\alpha.$$

Equivalently,

$$2^{(1-\alpha)H_\alpha(P)} = \sum_i p_i^\alpha 2^{(1-\alpha)H_\alpha(q_i)}.$$

This verifies the displayed identity in the track specification, with the important correction that its operadic observable is M_t^α , not M_t .

Desk-proved. The optimizer itself is compatible with hierarchical grafting only after the inner values reweight the outer problem. Optimize each inner share as

$$s_j^i = \frac{(q_j^i)^\alpha}{Z_i}.$$

The remaining outer objective is

$$\sum_i p_i R_i^{-t} M_t(q_i).$$

Its escort optimum, applied to the coefficients $p_i M_t(q_i)$, is

$$R_i = \frac{(p_i M_t(q_i))^\alpha}{\sum_k (p_k M_t(q_k))^\alpha} = \frac{p_i^\alpha Z_i}{Z_\alpha(P)}.$$

Therefore

$$R_i s_j^i = \frac{p_i^\alpha (q_j^i)^\alpha}{Z_\alpha(P)},$$

which is exactly the global escort share for P . Naively grafting the outer escort of p with the inner escorts of the q_i would omit the factors Z_i and is generally not globally optimal.

4.3 Raw cumulative machine travel

For level prices $c_t(\ell) = 2^{t\ell}$, cumulative travel to distance d is

$$C_t(d) = \sum_{\ell=0}^d 2^{t\ell} = \frac{2^t}{2^t - 1} 2^{td} - \frac{1}{2^t - 1}.$$

Desk-proved. Expected cumulative travel is a positive affine transform of the endpoint moment:

$$T_t(p, d) = \frac{2^t}{2^t - 1} E_t(p, d) - \frac{1}{2^t - 1}.$$

Thus it has the same continuous optimizer. Its normalized endpoint observable

$$G_t(T) = \frac{(2^t - 1)T + 1}{2^t}$$

is M_t at the optimum, and $G_t(T)^\alpha = Z_\alpha$. Raw travel itself does not satisfy (G); the affine normalization and α th power are essential.

4.4 What remains exact for integer prefix paths

Verified. Campbell's coding theorem gives the integer prefix-code bounds

$$H_\alpha(p) \leq \frac{1}{t} \log_2 E_t(p, d) < H_\alpha(p) + 1,$$

and block coding reduces the per-symbol gap to $1/m$. The full derivation and the corresponding affine raw-travel bounds are in `campbell-renyi.md` §2-3.

Wall, exact physical optimum. A continuous escort share is generally not a power of two, hence not the cylinder mass of one physical mount path. Moreover, an arbitrary globally optimal prefix code for the leaves (i, j) need not factor through a prescribed outer grouping i . Therefore (G) and (M) are exact for the continuous relaxation, while (F_t) is exact for each grafted integer mounting. There is no exact identity (M) for unrestricted one-letter integer mounting optima without changing the representation or admitting rounding slack.

5. Shannon is the derivative at the undeformed point

Use base-two logarithms. For $p_i > 0$,

$$\frac{1}{\ln 2} \frac{d}{d\alpha} Z_\alpha(p)|_{\alpha=1} = \sum_i p_i \log_2 p_i = -H(p).$$

Desk-proved. Differentiate (G) at $\alpha = 1$ and divide by $\ln 2$. Since $Z_1(q_i) = 1$,

$$\begin{aligned} -H(P) &= \sum_i p_i \log_2 p_i + \sum_i p_i (-H(q_i)) \\ &= -H(p) - \sum_i p_i H(q_i). \end{aligned}$$

Negating yields

$$H(p \circ (q_i)) = H(p) + \sum_i p_i H(q_i).$$

The same result follows by taking $\alpha \rightarrow 1$ in (T), since $S_\alpha \rightarrow H$ and $p_i^\alpha \rightarrow p_i$.

Verified. This is Faddeev's strong-additivity law and Bradley's scalar operadic Leibniz rule. Bradley proves that the constant function $d_p(x) = H(p)$ defines a derivation of the operad of topological simplices with values in an endomorphism-operad bimodule.

6. The full-simplex converse already exists

6.1 From an unknown scalar weight to a power

The literature usually fixes the deformation order. The following elementary step explains why a coordinatewise weight system must be a power when the order is initially unknown.

Desk-proved, conditional proposition. Suppose a nonzero symmetric family S_n on all finite probability simplices satisfies (D), including operadic units. Assume $w(0) = 0$, $w(x) > 0$ for $x > 0$, and w is monotone. Then

$$w(xy) = w(x)w(y) \quad (0 < x, y \leq 1).$$

To prove this, compare the two parenthesizations of a three-level refinement. Associativity and two applications of (D) give

$$\sum_{i,j} (w(p_i q_j^i) - w(p_i)w(q_j^i)) S(r^{ij}) = 0.$$

Choose all r^{ij} to be the one-point distribution except one, and choose the exception to have nonzero S . Since arbitrary $x, y \in (0, 1)$ can occur as coordinates of binary distributions, the surviving coefficient gives $w(xy) = w(x)w(y)$. The unit law gives $w(1) = 1$.

Put $x = e^{-s}$ and $a(s) = -\log w(e^{-s})$. Then $a(s+t) = a(s) + a(t)$ and monotonicity rules out the pathological Cauchy solutions, so $a(s) = \alpha s$. Consequently

$$w(x) = x^\alpha.$$

If w is nondecreasing with $w(0) = 0$, then $\alpha > 0$. Continuity can replace monotonicity here; monotonicity is the weaker usable regularity assumption.

6.2 Furuichi and Baez–Fritz–Leinster finish the entropy part

Verified. Furuichi's Theorem V.2 is a generalized Faddeev theorem: symmetry, continuity, and the recursion

$$S(tp_1, (1-t)p_1, p_2, \dots) = S(p_1, p_2, \dots) + p_1^\alpha S(t, 1-t)$$

characterize a nonnegative scalar multiple of order- α Tsallis entropy.

Verified. Baez, Fritz, and Leinster package this as their Theorem 7. A continuous functorial information-loss map compatible with convex sums by

$$F(\lambda f \oplus (1-\lambda)g) = \lambda^\alpha F(f) + (1-\lambda)^\alpha F(g)$$

is exactly a nonnegative multiple of Tsallis information loss. Their proof explicitly imports Furuichi. Applied to collapse maps, the statement is the p_i^α strong-additivity law (T).

Desk-proved. Rényi entropy is a monotone transform of the same object:

$$H_\alpha^R(p) = \frac{1}{1-\alpha} \log_2 Z_\alpha(p) = \frac{1}{1-\alpha} \log_2 (1 + (1-\alpha)S_\alpha(p)) .$$

Thus the operadic uniqueness theorem selects Tsallis in additive/exchange coordinates and selects Rényi after the logarithmic coordinate change. It does not make Rényi itself satisfy the linear deformed derivation law.

6.3 What Bradley proves

Verified, correction. Bradley defines derivations of an operad with values in an abelian bimodule and proves:

- the constant assignment $p \mapsto H(p)$ is a derivation of the topological simplex operad;
- for every such derivation d , evaluation at the zero vector satisfies $d_p(0) = cH(p)$ for some constant c .

The second statement relies on a Faddeev–Leinster characterization. It is weaker than saying every function-valued derivation d_p is itself constant at $cH(p)$. For this track, the scalar evaluation is enough to recover Shannon’s law on the full simplex, but Bradley alone does not provide the deformed Tsallis theorem; Furuichi and Baez–Fritz–Leinster do.

7. Conditional converse from grafting to exponential schedules

The entropy theorem classifies functionals on workloads. To classify machine prices, one must add a bridge from depth addition to the functional. There are two clean bridges.

7.1 Discrete affine-multiplicative endpoint costs

Desk-proved. Let $C : \mathbb{N} \rightarrow \mathbb{R}$ be endpoint or cumulative travel cost. Suppose there are constants $A \neq 0$ and B such that

$$G(d) = AC(d) + B > 0, \quad G(0) = 1,$$

and grafting depth satisfies

$$G(d + e) = G(d)G(e).$$

Set $q = G(1)$. Induction using (S) gives

$$G(d) = q^d, \quad C(d) = \frac{q^d - B}{A}.$$

If C is nonconstant and increasing, then $q > 1$. Its level increments are

$$c(d) = C(d) - C(d-1) = \frac{q-1}{A} q^{d-1} \quad (d \geq 1),$$

so all positive-depth level increments are geometric. The level-zero touch is an affine-baseline convention; the usual machine normalization below puts it on the same geometric progression. For the machine schedule $c_t(\ell) = 2^{t\ell}$,

$$C_t(d) = \frac{2^{t(d+1)} - 1}{2^t - 1}, \quad G(d) = \frac{(2^t - 1)C_t(d) + 1}{2^t} = 2^{td}.$$

This converse is purely discrete and needs no density argument. Its load-bearing axiom is the affine-normalized multiplicativity (S), not merely monotonicity or cumulative doubling.

7.2 A smooth optimizer converse

Desk-proved, conditional proposition. Let $f : (0, 1] \rightarrow \mathbb{R}$ be continuously differentiable and strictly decreasing. For every positive finite probability vector p , suppose the unique minimizer of

$$\sum_i p_i f(r_i) \quad \text{subject to} \quad r_i > 0, \quad \sum_i r_i = 1$$

is

$$r_i = \frac{p_i^\alpha}{\sum_j p_j^\alpha}$$

for one fixed $0 < \alpha < 1$. Then

$$f(r) = Ar^{-(1-\alpha)/\alpha} + B$$

for constants $A > 0$ and B .

Indeed, take any interior share vector r and choose $p_i \propto r_i^{1/\alpha}$, so that r is its assumed optimizer. The Lagrange stationarity equations give $p_i f'(r_i) = \lambda$, hence

$$r_i^{1/\alpha} f'(r_i) = r_j^{1/\alpha} f'(r_j)$$

for every pair of coordinates. By embedding arbitrary $x, y \in (0, 1)$ into interior probability vectors (using an intermediate coordinate if $x + y \geq 1$), $r^{1/\alpha} f'(r)$ is one constant on $(0, 1)$. Strict decrease makes it negative. Integration yields the displayed form.

Writing $r = 2^{-d}$ and $t = (1 - \alpha)/\alpha$ gives

$$C(d) := f(2^{-d}) = A2^{td} + B.$$

The limiting case $\alpha = 1$ yields $f(r) = -A \log_2 r + B$ and therefore $C(d) = Ad + B$, the constant-level-price Shannon case.

Wall, scope of converse. Neither proposition says that symmetry, monotonicity, and Kraft admissibility alone force exponentials. Section 7.1 assumes a graft-multiplicative affine normalization; §7.2 assumes a separable, differentiable continuous relaxation and a universal escort optimizer. The doubling axiom from `campbell1-renyi.md` §7.3 supplies integer rounding stability but does not imply either bridge.

8. Why the physical suboperad blocks the desired import

The full-simplex proof in §6 uses arbitrary binary coordinates twice:

1. associativity isolates $w(xy) = w(x)w(y)$ for arbitrary x, y ;
2. Faddeev's recursion varies t continuously and then uses continuity.

Wall. The complete physical two-head mounting image supplies only $x = t = 1/2$. There is no sequence of physical two-head mass vectors approaching, say, $(1/3, 2/3)$. Therefore neither use can be recovered from density.

Desk-proved. Replacing physical masses by all dyadic simplex points would repair the density step. If a family S and its grafting law are defined on $D_2(n)$ and extend continuously to $\Delta(n)$, then both sides of the law are continuous and equality extends from the dense dyadic subset. The full-simplex Furuichi/Baez–Fritz–Leinster theorem then applies.

Wall, machine meaning. That repair is not free. A dyadic coordinate $k/2^N$ represents a finite union of cylinders, not one mount path; a head no longer has one distance. Alternatively, one can keep physical mountings and put the simplex operad on workload vectors p . Then Campbell optimization is a map from workload distributions to *continuous* shares, with integer mountings as rounding approximations. This second separation preserves the machine model but makes the operadic classification a theorem about the relaxation, not an exact classification internal to the mounting operad.

Literature check, no located import. Targeted searches around Faddeev, simplex operads, dyadic probabilities, and restricted/dyadic suboperads found the full-simplex characterizations and algebraic constructions involving Rényi/Tsallis, but no theorem that derives the full continuous classification from the one-cylinder-per-coordinate image $P_2(n)$. This is a report of the search performed, not a proof that no such result exists.

9. What exactly is canonical

The chain of implications that survives is:

full workload simplex + continuous generalized grafting law

$\Downarrow$ Furuichi / Baez–Fritz–Leinster

Tsallis S_α and weights p_i^α

$\Downarrow$ $Z = 1 + (1 - \alpha)S_\alpha$

$Z_\alpha(p) = \sum_i p_i^\alpha$

$\Downarrow$ monotone logarithm

Rényi H_α

$\Downarrow$ separable Campbell optimization

$r_i \propto p_i^\alpha, \quad M_t = Z_\alpha^{1/\alpha}, \quad t = (1 - \alpha)/\alpha.$

Verified/desk-proved. The first implication is imported; the remaining algebra and optimization are desk-proved above. With either schedule bridge in §7, the endpoint cost is exponential and the cumulative level schedule is geometric.

Wall. Without a schedule bridge, an entropy characterization does not classify arbitrary machine costs. Without broadening or separating the physical mounting masses, their suboperad is too small for the full-simplex converse. The operadic line therefore sharpens the canonicity claim substantially but does not make it unconditional.

10. Workaround: couple workloads to mountings instead of identifying them

The density wall is avoidable. It came from trying to make one vector do two jobs. Workload probabilities already range over a simplex; physical mount masses do not need to.

10.1 The scheduled-mounting operad

Definition. Let

$$\mathcal{A}(n) = \Delta(n) \times \mathcal{M}(n).$$

An element (p, u) is a workload distribution together with a physical mounting of the same arity. Compose componentwise:

$$(p, u) \circ ((q_i, v^i)_i) = (p \circ (q_i), u \circ (v^i)) .$$

Desk-proved. Since both factors are symmetric operads, their aritywise product is a symmetric operad. Its composite has workload coordinates $p_i q_j^i$ and physical depths $d_i + e_j^i$. No map from Kraft masses to workload probabilities is asserted or needed.

For a normalized endpoint price $g : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ with $g(0) = 1$, define the separable evaluation

$$E_g(p, u) = \sum_i p_i g(d_i(u)).$$

This is an observable on the *paired* operad, not a functional on either factor alone.

10.2 Operadic locality forces the exponential

The appropriate local grafting axiom says that the contribution of outer slot i to a refined schedule depends only on its own probability and depth.

Axiom (local factorization). There is a scalar coefficient $a : [0, 1] \times \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ such that for every scheduled mounting and every family grafted below it,

$$E_g(p \circ (q_i), u \circ (v^i)) = \sum_i a(p_i, d_i) E_g(q_i, v^i).$$

Desk-proved, exact converse. For every coefficient pair that occurs in a scheduled mounting, Axiom (L) forces

$$a(p, d) = p g(d), \quad g(d + e) = g(d)g(e),$$

and hence $g(d) = q^d$ for $q = g(1)$.

First restrict (L) to unary outer and inner mountings. Their workload probability is one, while their paths may have arbitrary depths d and e . The law says

$$g(d + e) = a(1, d)g(e).$$

Putting $e = 0$ gives $a(1, d) = g(d)$, hence $g(d + e) = g(d)g(e)$. Induction gives $g(d) = q^d$. If g is increasing and nonconstant, then $q > 1$.

It remains to identify the general coefficient. Once $q \neq 1$, fix an outer operation and choose unary inner paths of depths e_i . Equation (L) becomes

$$\sum_i p_i g(d_i) q^{e_i} = \sum_i a(p_i, d_i) q^{e_i}.$$

First set every $e_i = 0$, then change only e_i to one and subtract the two real equalities. Since $q - 1 \neq 0$, this isolates $a(p_i, d_i) = p_i g(d_i)$. The constant case $g \equiv 1$ is the trivial schedule; (L) then determines only the sum of the coefficients and is intentionally excluded from the nontrivial classification.

This theorem is stronger and cleaner than attempting to recover exponential prices from density. It uses only literal integer paths and literal grafting. For cumulative travel C , apply it to an affine normalization $g(d) = AC(d) + B$ as in §7.1; the level increments of C are then geometric.

Desk-proved. Conversely, $g(d) = q^d$ satisfies (L) exactly, with $a(p, d) = pq^d$, by (F_t) . Thus local factorization characterizes exponential endpoint schedules on the physical paired operad, rather than merely showing they are examples.

Desk-proved, undeformed branch. The Shannon endpoint observable uses the additive character $h(d) = d$, not the multiplicative character $g(d) = q^d$. The corresponding local law is

$$E_h(p \circ (q_i), u \circ (v^i)) = E_h(p, u) + \sum_i p_i E_h(q_i, v^i).$$

Conversely, testing a normalized endpoint price $h(0) = 0$ on unary grafts forces $h(d + e) = h(d) + h(e)$, hence $h(d) = d h(1)$ by induction. Its cumulative travel is affine in $1 + d$, so its level increments are constant. It is also the regular limit

$$\lim_{t \rightarrow 0} \frac{2^{td} - 1}{2^t - 1} = d.$$

Thus the paired operad classifies both branches: additive/linear gives Shannon, while multiplicative/nonconstant gives the positive- t exponential family.

10.3 Fiberwise optimization produces the deformed simplex law

Relax the mounting fiber over a workload p from integer paths to complete real shares $r_i > 0$, $\sum_i r_i = 1$, with $d_i = -\log_2 r_i$. For $q = 2^t$, minimize the paired observable along that fiber:

$$\mathcal{V}_t(p) = \inf_r \sum_i p_i r_i^{-t}.$$

Desk-proved. Section 4 gives

$$\mathcal{V}_t(p) = Z_\alpha(p)^{1/\alpha}, \quad \alpha = \frac{1}{1+t},$$

and therefore

$$\mathcal{V}_t(p \circ (q_i))^\alpha = \sum_i p_i^\alpha \mathcal{V}_t(q_i)^\alpha.$$

So the p_i^α law is not postulated on physical Kraft masses. It is the law induced on the workload factor by fiberwise optimization of the uniquely graft-local physical price family. The three canonicity steps are now cleanly separated:

$$\begin{aligned} & \text{local factorization on } \Delta \times \mathcal{M} \\ & \Downarrow \\ & g(d) = 2^{td} \text{ (or } h(d) = d \text{ in the additive branch)} \\ & \Downarrow \text{ continuous Kraft minimization} \\ & r_i \propto p_i^{1/(1+t)}, \quad Z_\alpha = \mathcal{V}_t^\alpha \\ & \Downarrow \text{ Furuichi / BFL} \\ & \text{the induced continuous grafting functional is Tsallis,} \\ & \text{or Rényi after the logarithmic transform.} \end{aligned}$$

Desk-proved, workaround verdict. This coupled-operad theorem pins the classification within a natural and explicit axiom set while leaving the one-path-per-head machine representation untouched. The remaining relaxation gap is Campbell's familiar integer rounding gap, not an operadic density gap. For the paper, this is the recommended repair.

10.4 Optional extension: the labelled dyadic envelope

If a literal dense dyadic suboperad is desirable for other reasons, there is a second, representation-enlarging workaround.

Definition. A labelled dyadic operation of arity n consists of a finite complete prefix code $L \subset B^*$ and a labelling $\lambda : L \rightarrow \{1, \dots, n\}$, not necessarily surjective. Its mass vector is

$$p_i = \sum_{u \in L: \lambda(u)=i} 2^{-|u|}.$$

Graft a copy of the inner labelled code for i below every outer leaf labelled i , and retain the inner label on the composite leaf.

Desk-proved. These operations form a symmetric operad. Their mass map lands in $D_2(n)$ and is surjective: for $p_i = k_i/2^N$ with $\sum_i k_i = 2^N$, take all 2^N words of depth N and label exactly k_i of them by i . Grafting multiplies and sums masses, giving exactly $(p_i q_j)_{i,j}$. Hence this extension

has a dense mass image. A uniformly continuous mass-local law on this dyadic subspace extends uniquely to the full simplex; alternatively, continuous extendability can be stated as the regularity axiom. Mere continuity on a non-complete dense subspace is not enough.

Wall, physical interpretation. A label may occur on many leaves, so this is not a static one-path-per-head mounting. It can model a dyadic randomized router, a coarse-graining of mounted micro-outcomes, or a block-code relaxation. Adopting it as literal fast-state geometry would require a rule for choosing among replicas and maintaining coherent head state. It is therefore a useful algebraic envelope, but not the recommended core-machine change.

11. Replacement for `campbell-renyi.md` §8

The qualified claim in that report can be replaced by the following precise statement:

Verified plus desk-proved. On the full operad of finite probability simplices, a continuous nontrivial symmetric grafting functional with a monotone coordinatewise weight system has power weights p_i^α ; Furuichi’s generalized Faddeev theorem, equivalently the Baez–Fritz–Leinster information-loss theorem, then makes the functional a scalar Tsallis entropy. Its positive partition coordinate is $Z_\alpha = \sum_i p_i^\alpha$, its logarithmic coordinate is Rényi entropy, and Campbell’s separable Kraft optimization has escort shares and endpoint value $Z_\alpha^{1/\alpha}$. If affine-normalized endpoint price is multiplicative under additive depths—or, in the smooth relaxation, if the universal optimizer is escort—then the machine schedule is exponential. The repository’s literal one-path-per-head mounting masses are not dense in the simplex, so this converse is a theorem of the workload/continuous-share layer, not of the physical mounting suboperad alone.

This is stronger than the earlier appeal to Rényi’s quasi-arithmetic means: the deformed grafting uniqueness is a direct imported theorem, and the exact extra hypothesis needed to reach machine schedules is visible.

The preferred formulation can now be sharper still: use the scheduled-mounting operad $\Delta \times \mathcal{M}$. Local factorization on its physical depth component forces the exponential schedule exactly; fiberwise Kraft optimization induces the escort/Tsallis/Rényi law on its workload component. No density claim about physical Kraft masses is needed.

12. Lean mechanization ladder

Recommendation (design, not decision). Mechanize in separate namespaces so the physical and continuous layers cannot be conflated.

1. **Bit-word operad (easy).** Define mounting graft by list append; prove prefix-incomparability, associativity, units, permutation compatibility, `graft_depth`, and `graft_kraft_mass`. Reuse `Mounting` and `PrefixIncomparable` from `Adic.Distance/Adic.Locality`.
2. **Physical-image obstruction (easy).** Prove that tight two-head distances satisfy $d_0 = d_1 = 1$, and record that the physical image is not the dyadic simplex. This is all-natural power arithmetic.

3. **Fixed-mounting graft costs (easy/medium)**. For natural macro prices prove the analogues of (F) and (F_t) by finite-sum rearrangement and `Nat.pow_add`.
4. **Partition operad algebra (medium)**. Over `NNReal` or `Real`, define simplex composition and Z_α for a rationally represented order; prove (G), the Tsallis exchange identity, and the Rényi logarithmic transform.
5. **Campbell optimum (hard)**. Reuse the proposed `Adic.Campbell` ladder: denominator-cleared natural Hölder first, then connect it to the real escort optimum and $M_t^\alpha = Z_\alpha$. This remains the hard analytic rung already named in `campbell-renyi.md`.
6. **Shannon derivative (medium/hard)**. Prove differentiability of finite power sums at $\alpha = 1$ and differentiate (G), or avoid calculus by proving the Shannon chain rule directly and separately proving the Tsallis limit.
7. **Weight-system converse (hard)**. Formalize the associativity isolation argument, monotone multiplicative functions on $(0, 1]$, and the reduction to Furuichi. The final uniqueness theorem is not currently in Mathlib and would require a substantial functional-equation development or be stated as an imported paper theorem rather than immediately mechanized.
8. **Paired-operad schedule converse (easy/medium)**. Define the product of workload vectors and mountings, state local factorization (L), isolate one summand, and derive $g(d + e) = g d * g e$ and $g d = (g 1) ^ d$. This is the recommended first canonicity theorem.
9. **Smooth schedule converse (hard)**. The escort converse needs constrained first-order optimality and integration over reals.
10. **Optional labelled envelope (medium)**. Define complete prefix codes with repeated finite labels, graft them, prove the mass composition law, and construct a labelled full-depth code for every dyadic simplex point.

The first three rungs certify the exact machine-native content. Rungs 4–6 certify the forward relaxation. Rungs 7–8 certify the conditional canonicity claim. None should state that physical mounting masses are dense.

13. Open items

With Mathijs

1. Should the paper adopt the **scheduled-mounting product operad** $\Delta \times \mathcal{M}$ (recommended), or also present the labelled dyadic envelope as a secondary algebraic completion?
2. Is affine-normalized multiplicativity under depth addition accepted as the schedule-canonicity axiom, or should the paper instead use the smoother but stronger universal-escort-optimizer axiom?
3. Should the exact operadic observable be named the *partition functional* $Z_\alpha = M_t^\alpha$, reserving M_t for Campbell’s endpoint moment (recommended)?

For agents

1. Add the small mounting-graft API, the product-operad evaluation, and the local-factorization schedule converse to Lean.
2. Mechanize the two-head physical-image obstruction before any prose calls the image “the dyadic simplex”.

3. If the full converse is prioritized, first survey Mathlib for monotone multiplicative-function and Faddeev-style functional-equation support; do not begin from Bradley’s theorem as though it classified the entire function-valued derivation.

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